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PRN: B25ME1027 Div: SY3 Class: C3

INPUT:

```
#include <iostream>
using namespace std;

class Shape
{
public:
   virtual double area() = 0;      // pure virtual function
   virtual ~Shape() {}             // virtual destructor
};

class Circle : public Shape
{
   double radius;
public:
   Circle(double r) { radius = r; }
   double area() override { return 3.14159265 * radius * radius; }
};

class Rectangle : public Shape
{
   double length, width;
public:
   Rectangle(double l, double w) { length = l; width = w; }
   double area() override { return length * width; }
};

int main()
{
   Circle c(3);
   Rectangle r(4, 5);

   Shape *s1 = &c;     // base class pointers
   Shape *s2 = &r;

   cout << "Area of circle    = " << s1->area() << endl;
   cout << "Area of rectangle = " << s2->area() << endl;
   return 0;
}
```

OUTPUT:

Output

```
Area of circle    = 28.2743
Area of rectangle = 20


=== Code Execution Successful ===
```